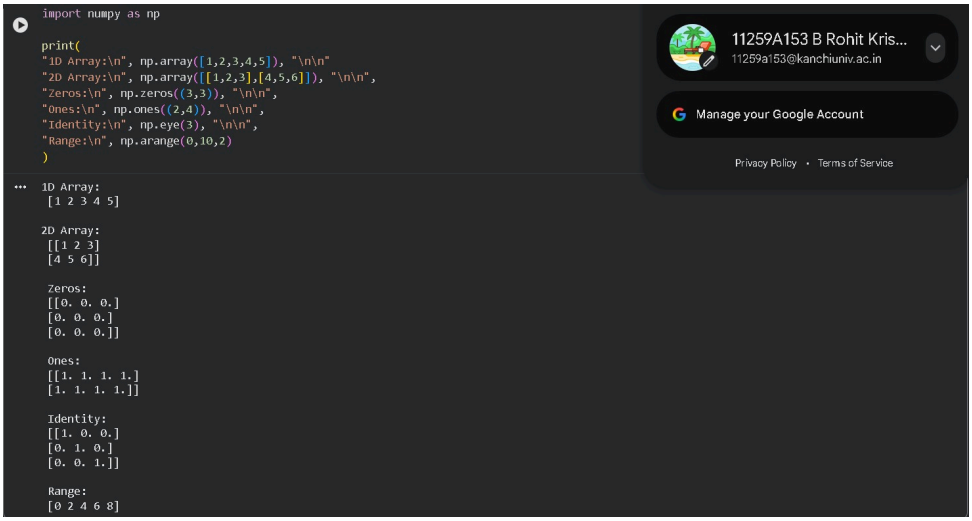
NumPy (Numerical Python)

A. Creating Arrays

AIM: To write a Python program using NumPy to create a one-dimensional array, two-dimensional array, zeros array, ones array, identity matrix, and range of values.

ALGORITHM:

1. Start the program.
2. Import the NumPy library.
3. Create a one-dimensional array using np.array().
4. Create a two-dimensional array using np.array().
5. Create a 3 × 3 zeros array using np.zeros().
6. Create a 2 × 4 ones array using np.ones().
7. Create a 3 × 3 identity matrix using np.eye().
8. Generate numbers from 0 to 8 with a step of 2 using np.arange().
9. Display all the arrays.
10. Stop the program.



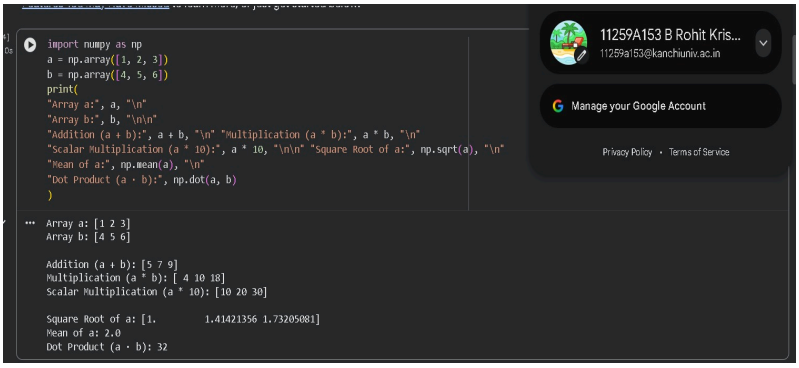
RESULT: Thus, the Python program to create one-dimensional and two-dimensional arrays, zeros array, ones array, identity matrix, and range of values using NumPy was executed successfully

B. Array Operations

AIM: To write a Python program to perform addition, multiplication, scalar multiplication, square root, mean, and dot product operations using NumPy arrays.

ALGORITHM:

1. Start the program.
2. Import the NumPy library.
3. Create two arrays, a = [1, 2, 3] and b = [4, 5, 6].
4. Perform element-wise addition using a + b.
5. Perform element-wise multiplication using a * b.
6. Multiply each element of array a by 10 using a * 10.
7. Calculate the square root of array a using np.sqrt().
8. Calculate the mean of array a using np.mean().
9. Calculate the dot product of arrays a and b using np.dot().
10. Display all the results.
11. Stop the program.



RESULT: Thus, the Python program to perform addition, multiplication, scalar multiplication, square root, mean, and dot product operations using NumPy arrays was executed successfully.

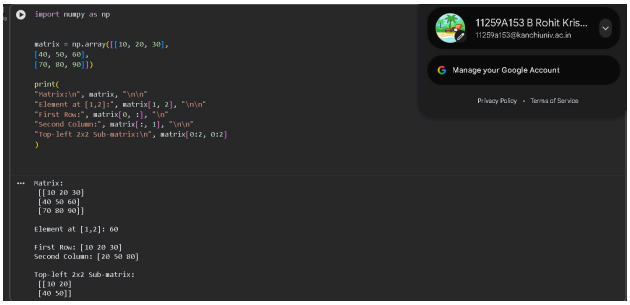
C. Indexing and Slicing

AIM: To write a Python program to access matrix elements, extract rows and columns, and obtain a sub-matrix using NumPy indexing and slicing.

ALGORITHM:

1. Start the program.

2. Import the NumPy library.
3. Create a 3 × 3 matrix containing the values 10, 20, 30, 40, 50, 60, 70, 80, and 90.
4. Display the complete matrix.
5. Access and display the element at index [1, 2].
6. Extract and display the first row using matrix[0, :].
7. Extract and display the second column using matrix[:, 1].
8. Extract and display the top-left 2 × 2 sub-matrix using matrix[0:2, 0:2].
9. Stop the program.



RESULT: Thus, the Python program to access matrix elements, extract rows and columns, and obtain a sub-matrix using NumPy indexing and slicing was executed successfully.